

TOC (CS302)

Unit-wise Syllabus Breakdown

Unit 1 – DFA & NFA

Study of finite automata, state transitions, deterministic finite automata (DFA), nondeterministic finite automata (NFA), and ϵ -NFA conversions.

Unit 2 – Regular Languages

Regular expressions, regular languages, closure properties, minimization of finite automata, and the pumping lemma.

Unit 3 – Context-Free Grammars

Context-free grammars (CFGs), derivations, parse trees, ambiguity, simplification techniques, and Chomsky Normal Form (CNF).

Unit 4 – Pushdown Automata

Stack-based automata, PDA design, acceptance conditions, and the equivalence between PDAs and CFGs.

Unit 5 – Turing Machines & Decidability

Turing Machines, decidability, recursive and recursively enumerable languages, reductions, and the halting problem.